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1 Summary

For this project, I evaluated the performance of three Large Language Model (LLM) architectures - Inference, Supervised Fine-Tuning (SFT) and Retrieval-Augmented Generation (RAG) - within the specialized domain of Austrian Tax Law (Österreichisches Steuerrecht).

2 Model Specification and Architecture

I used three distinct models, each selected for its specific performance-to-size ratio and reasoning capabilities.

Feature	**Inference Model**	**Fine-Tuned Model**	**RAG Model**
Model Name	GPT-4o-mini	Meta Llama 3.1 8B	Google Gemma 1.1 2B-it
Parameters	Undisclosed	8 Billion	2 Billion
Sampling	Zero-Shot (Temp 1.0)	Greedy Decoding: do_sample = False for reproducibility and precision.	Greedy Decoding, do_sample = False for reproducibility and precision.
Training Data	Austrian Legal Texts (public web/code)	15 Trillion tokens (public web/code) + ChatGPT generated samples about EStG, KStG, BAO and specialized tax laws to transition from a general LLM to an Austrian Tax Specialist.	PDF about Austrian Tax Law (BAO, KStG, EStG, UStG)
Implementation	Used as a baseline to evaluate pre-existing knowledge of technical German legal terminology and statutes like the EStG (Income Tax Act).	Used the Unsloth library to implement QLoRA (4-bit quantization), updating low-rank matrices while freezing base weights (15 trillion tokens of pre-trained knowledge) to reduce memory overhead.	Used ChromaDB as a vector store and sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 for semantic search.

2.1 Inference Model

For this project, I used GPT-4o-mini via the OpenAI API as the primary inference model. This model is a small-scale, high-performance variant of the GPT-4 family, specifically designed for tasks requiring a balance between speed, cost-efficiency and deep contextual understanding. GPT-4o-mini is a transformer-based model. While OpenAI does not publicly disclose the exact parameter count, it is optimized to provide reasoning capabilities comparable to larger models (OpenAI Developers, 2026).

I employed Zero-Shot Inference to evaluate the model's base knowledge of Austrian Tax Law, and used the default temperature (1.0) for processing.

Additionally, I prompted the inference model to be a "Expert for Austrian Tax Law." This instructed the model to prioritize specific legal terminology and verify that cited paragraphs corresponded correctly to categories like UStG (VAT) or KStG (Corporate Tax). The model was pre-trained on a vast, web-scale corpus of text and code.

2.2 Fine-Tuned Model

The fine-tuning was performed on Meta Llama 3.1 8B, an 8-billion-parameter large language model. During training, greedy decoding (do_sample = False) was used to ensure deterministic outputs, improving reproducibility and consistency across evaluations.

The model had originally been pre-trained on approximately 15 trillion tokens of publicly available web and code data, providing broad general-domain knowledge. This base capability was then adapted to the target domain using ChatGPT-generated training samples covering Austrian tax law, including the EStG, KStG, BAO, and other specialized tax regulations. The goal of this fine-tuning step was to transform the general-purpose model into a domain-specific Austrian tax assistant.

For parameter-efficient fine-tuning, I used the Unsloth library to implement QLoRA (Quantized Low-Rank Adaptation). This approach applies 4-bit quantization to the base model while freezing the original weights and training only low-rank adaptation matrices. This significantly reduces memory consumption and computational cost while preserving the capabilities of the underlying pre-trained model.

2.3 RAG Model

The retrieval-augmented generation (RAG) system is built on a specialized knowledge base derived from a PDF document (BAO, KStG 1988, EStG 1988, UStG 1994.pdf) focused on Austrian tax regulations. The document was segmented using a recursive character-based splitting strategy (RecursiveCharacterTextSplitter) into chunks of 500 characters with an overlap of 150 characters to ensure effective retrieval. This overlap was introduced to preserve semantic continuity across section boundaries and reduce information loss in context-sensitive legal passages.

The RAG generation model is based on Google Gemma 1.1 2B-it, a 2-billion-parameter instruction-tuned language model. Greedy decoding (do_sample = False) is used to ensure deterministic outputs and improve reproducibility and precision across evaluations. The system relies on ChromaDB as the vector database for efficient similarity search, while embeddings are generated using the sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 model, which enables multilingual semantic retrieval across Austrian legal terminology.

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3 Performance Evaluation (Main Result Table)

The performance was measured using **BERTScore** with German language settings (lang="de") to capture technical nuances. Unlike BLEU, BERTScore evaluates semantic alignment rather than exact word matching, which is critical for legal principles.

Since a shared ground-truth answer set was not available because dataset_clean.csv only contains id and prompt (and not id, prompt and correct_answer), I expanded dataset_clean.csv with the column "correct_answer" (dataset_clean_fixed.csv) to compare the model results against human-generated answers.

Model	BERTScore Precision	BERTScore Recall	BERTScore F1
Inference (GPT-4o-mini)	0.0537	**0.3087**	0.1718
Fine-Tuned (Llama 3.1 8B)	**0.3236**	0.2437	**0.2812**
RAG (Gemma 1.1 2B)	0.2706	0.1764	0.2211

The Fine-Tuned Model achieved the highest F1 BERTScore.

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4 Error Analysis

The following analysis details the specific failure modes observed during the evaluation of the baseline, fine-tuned, and RAG models.

Since automated evaluation of factual correctness in the legal domain requires a fully validated ground truth dataset, a manual spot-check was conducted. Ten representative prompts were selected from dataset_clean_fixed.csv, covering a range of Austrian tax law categories including VAT, corporate tax, income tax, and real estate transfer tax. Each model's generated answer was manually compared against the verified ground truth.

Prompt	**Correct Answer**
CORP-TAX-017: Was ist der Unterschied zwischen § 11 KStG und § 12 KStG in Bezug auf den Betriebsausgabenabzug?	§ 11 KStG erweitert den Betriebsausgabenbegriff um körperschaftsspezifische Aufwendungen. § 12 KStG schränkt ihn ein und nennt Aufwendungen, die trotz betrieblicher Veranlassung nicht abzugsfähig sind.
TAX-INTL-053: Ist für die beschränkte Steuerpflicht selbständiger Arbeit im Inland zwingend eine österreichische Betriebsstätte erforderlich?	Nein. Für Einkünfte aus selbständiger Arbeit ist eine Betriebsstätte nach innerstaatlichem Recht keine Voraussetzung; maßgeblich ist die Ausübung oder Verwertung der Arbeit im Inland.
VAT-DOM-056: Ein Verlag verkauft eine gedruckte Wochenzeitung, die zwar mehrere Anzeigen und Werbeseiten enthält, aber ansonsten als klassische periodische Druckschrift erscheint. Es wird ein einheitlicher Preis verlangt. Unterliegt die Lieferung dem 10%-Satz?	Ja. Werbliche Inhalte schließen die Begünstigung für Zeitungen und andere periodische Druckschriften nicht aus.
GRESt-AT-015: Ein Käufer und ein Verkäufer schließen einen Kaufvertrag über eine Eigentumswohnung ab. Wer ist gegenüber dem Finanzamt für die Zahlung der Grunderwerbsteuer verantwortlich?	Gemäß § 13 Abs. 1 GrEStG sind die am Erwerbsvorgang beteiligten Personen (Käufer und Verkäufer) Steuerschuldner. Sie haften als Gesamtschuldner. Das bedeutet, das Finanzamt kann sich die Steuer von jedem der beiden Vertragspartner in voller Höhe holen, auch wenn im Vertrag vereinbart wurde, dass der Käufer die Steuer trägt.

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VAT-INTL-074: Ein Maler aus Frankreich vermietet seine Gemälde an Museen in Österreich, die die Werke für eine Ausstellung nutzen. Wo ist die Leistung steuerpflichtig?	In Österreich. Es handelt sich um eine B2B-Leistung. Gemäß § 3a Abs. 6 UStG 1994 wird eine sonstige Leistung, die an einen Unternehmer im Sinne des Abs. 5 Z 1 und 2 ausgeführt wird, vorbehaltlich der Abs. 8 bis 16 und Art. 3a an dem Ort ausgeführt, von dem aus der Empfänger sein Unternehmen betreibt. Wird die sonstige Leistung an die Betriebsstätte eines Unternehmers ausgeführt, ist stattdessen der Ort der Betriebsstätte maßgebend.
ESTG27-036: Frau Hahn hat Aktien der Z-AG in ihrem Depot bei der österreichischen Bank A. Sie beauftragt Bank A, alle Aktien auf ihr neues Depot bei derselben Bank A zu übertragen (interner Depottransfer). Einige Monate später überträgt sie dieselben Aktien auf ihr Depot bei der österreichischen Bank B, wobei sie Bank A beauftragt, Bank B die Anschaffungskosten mitzuteilen. Lösen diese Vorgänge steuerliche Konsequenzen nach § 27 EStG 1988 aus?	Nein, keiner der beiden Vorgänge löst steuerliche Konsequenzen aus. Gemäß § 27 Abs. 6 Z 2 EStG 1988 liegt in folgenden Fällen keine Veräußerung vor: (1) Bei der Übertragung auf ein anderes Depot desselben Steuerpflichtigen bei derselben depotführenden Stelle (interner Depottransfer bei Bank A) – hier besteht keine Melde- oder Beauftragungspflicht. (2) Bei der Übertragung auf ein Depot desselben Steuerpflichtigen bei einer inländischen depotführenden Stelle (Bank B), wenn der Steuerpflichtige die übertragende depotführende Stelle (Bank A) beauftragt, der übernehmenden Stelle (Bank B) die Anschaffungskosten mitzuteilen. Beide Vorgänge erfüllen die gesetzlichen Voraussetzungen, sodass keine Veräußerung und damit keine Steuerpflicht entsteht. Die Anschaffungskosten werden bei Bank B fortgeführt.
EMP-007: Ein Außendienstmitarbeiter erhält für eine Dienstreise Taggelder in Höhe von 26,40 € pro Tag. Er fragt sich, ob er diesen Betrag in seiner Steuererklärung als Einkommen versteuern muss.	Nein. Soweit die Vergütungen für Dienstreisen die Sätze des § 26 EStG nicht übersteigen (z.B. das amtliche Kilometergeld oder Taggelder bis 26,40 €), gehören sie nicht zu den steuerpflichtigen Einkünften aus nichtselbständiger Arbeit.
EStG-23-040: Ein Bauer betreibt zusätzlich zu seiner Landwirtschaft einen kleinen Hofladen, in dem er eigene Produkte verkauft. Er meint, da er Landwirt ist, falle alles unter Land- und Forstwirtschaft und nicht unter § 23. Stimmt das?	Nein. Da der Bauer ausschließlich eigene Erzeugnisse vermarktet, bleibt die Tätigkeit im Rahmen der Land- und Forstwirtschaft. § 23 Z 1 schließt Land- und Forstwirtschaft ausdrücklich aus, ein Gewerbebetrieb liegt nicht vor.

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SELF-023: Unterliegt ein Import eines PKW aus der EU der NoVA?	Ja, bei der erstmaligen Zulassung im Inland ist die Normverbrauchsabgabe fällig.
SELF-065: Was ist der Inhalt des Progressionsabgeltungsgesetzes 2025?	Die Anpassung der Grenzbeträge des Einkommensteuertarifs an die Inflation (Abschaffung kalte Progression).

Prompt-ID:	**Correct Answer (Ground Truth)**	**Inference Model (Baseline)**	**Fine-Tuned Model (SFT)**	**RAG Model (Retrieval)**
CORP-TAX-017: (§ 11 KStG 1988; § 12 KStG 1988)	§ 11 expands (corp-specific); § 12 restricts (exclusions).	Correct. Correctly explains § 11 (expands) and § 12 (restricts) with correct citations, but misses the key detail that § 11 expands specifically for corporation-specific expenditures, and is very verbose.	Partly Correct. Correctly cites § 11 KStG and § 12 KStG, but describes § 12 as covering "Sonderausgaben like insurance premiums" which is not in the ground truth, missing that § 12 restricts business expenses that are otherwise operationally motivated. Minor terminological drift.	Partly Correct. Correctly cites § 11 KStG and § 12 KStG, but narrows the entire difference down to donations to specific institutions only, completely missing the general principle that § 11 expands and § 12 restricts business expense deductibility. The reference to Abs. 6 Z 2 and 12 is not in the ground truth.
TAX-INTL-053 (EStR 2000 Rz 7914)	No. For income from self-employed work, a permanent establishment under domestic law is not a prerequisite; what matters is whether the work is performed or exploited domestically. (Cites no specific law)	Partly Correct. Correctly concludes no permanent establishment is required, but cites § 1 Abs. 3 EStG (which governs unlimited tax liability, not limited tax liability) and adds three unreferenced conditions not in the ground truth.	Incorrect. Claims a foreign permanent establishment suffices, missing that no establishment is required at all. Cites § 3 Abs. 2 EstG, which is unrelated to the question.	Fail. No relevant document found in the knowledge base - question left unanswered.

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VAT-DOM-056 (§ 10 Abs. 2 Z 1 lit. a UStG 1994; Anlage 1 Z 33 lit. b UStG 1994)	Yes. Advertising doesn't exclude the 10% rate for periodicals. (Cites no specific law)	Partly Correct. Reaches the correct conclusion (10% applies) but adds a condition not in the ground truth - that advertising must be "not dominant" for the rate to apply. The ground truth explicitly says advertising content does not exclude the benefit at all. Cites § 10 Abs. 2 Z 1 UStG which is the right provision. Verbose.	Incorrect. Answers "No" and wrongly claims the delivery is VAT-exempt under § 6 Abs. 1 Z 1 UStG - a completely wrong law and wrong conclusion.	Partly Correct. Reaches the correct conclusion (10% applies) but gives no legal basis and vague reasoning. Doesn't mention that advertising content is irrelevant, just states the rate applies.
GREStG-AT-015 (§ 13 Abs. 1 GrEStG 1987)	Both buyer and seller are liable as joint/several debtors (§ 13 Abs 1. GrEStG).	Incorrect. Wrongly claims only the buyer is liable, and cites the wrong paragraph (§ 2 and § 1 GrEStG instead of § 13 Abs 1. GrEStG).	Inorrect. Wrongly claims only the buyer is liable, and cites the wrong tax law (§ 27 Abs. 1 UStG instead of § 13 Abs 1. GrEStG).	Incorrect. Wrongly claims only the seller is liable, and does not cite the legal basis.
VAT-INTL-074 (§ 3a Abs. 6 UStG 1994)	In Austria (B2B rule, § 3a Abs 6).	Partly Correct. Confirms AT taxability but cites § 3 Abs 8 UStG instead of § 3a Abs. 6 UStG.	Partly Correct. Identifies Austrian taxability correctly and cites § 3a UStG - right tax law but no specific paragraph.	Incorrect. Claims the painter is "taxable in Austria" but without any concrete legal basis.
ESTG27-036 (§ 27 Abs. 6 Z 2 EStG 1988)	No tax consequences for internal/domestic depot transfers (§ 27 Abs 6).	Partly Correct. Accurately describes that transfers do not trigger realization of gains, but cites § 27 Abs. 1 instead of the specific § 27 Abs. 6 Z 2.	Incorrect. Correctly says "No" (§ 27 EstG) but invokes restructuring law (§ 2 UmgrStG) which is completely irrelevant.	Incorrect. Inverts the conclusion - claims transfers are "not permissible" when ground truth says no tax consequences arise. Fabricates a rule about acquisition costs not being accounted for, with no legal citation.

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EMP-007 (§ 26 Z 4 EstG)	No. Tax-free if daily allowances are <= €26.40 (§ 26 EStG).	Partly Correct. Reaches the correct conclusion (tax-free up to €26.40) and correctly identifies the threshold. However cites § 16 Abs. 1 Z 8 EStG (Werbungskosten/business expenses) instead of the correct § 26 EStG - and adds an unreferenced 12-hour absence condition not in the ground truth.	Incorrect. Correctly says "No" but for the wrong reason - claims daily allowances are tax-free as Sonderausgaben under § 18 Abs. 1 Z 3 EStG.	Incorrect. States it must be taxed but hourly rates are free. § 16d Abs. 1 Nummer 1 des österreichischen Steuergesetzes - both the paragraph and "österreichisches Steuergesetz" are fabricated, neither exists in Austrian tax law.
EstG-23-040 (§ 23 EStG 1988)	No. Farming excludes § 23 (remains Land/Forst).	Incorrect. Wrongly concludes the farm shop falls under § 23 EStG as commercial income, directly contradicting the ground truth. Introduces irrelevant UStG considerations and a revenue threshold that has no basis in the question. Cites non-existent § 34 Abs. 1 EStG.	Incorrect. Correctly says "No" but for the completely wrong reason - instead of addressing whether the farm shop falls under § 23 EStG or agriculture, it pivots entirely to VAT law and cites § 3a UStG (place of supply rules for services), which is completely irrelevant to the question.	Incorrect. Wrongly agrees with the farmer that everything falls under agriculture. Cites § 21 Abs. 3 Z 1 der Steuergesetz - a fabricated provision, as § 21 EStG only has two paragraphs and "der Steuergesetz" is not a real legal reference.
SELF-023 (NoVAG § 1)	Yes. NoVA is due on first domestic registration of an EU car.	Correct. Correctly concludes NoVA is due on first registration in Austria. Correctly cites § 1 Abs. 1 NoVAG and § 6 NoVAG. Adds CO_2 calculation detail not in ground truth but relevant and accurate.	Incorrect. Answers "No" and claims NoVA does not apply because "the business already exists" - completely fabricated reasoning. Cites § 6 Abs. 1 Z 1 UStG (VAT exemptions) which is entirely unrelated.	Incorrect. It says NoVA does not apply, then immediately says the law provides no exemptions for EU cars, which would imply NoVA does apply. The RAG model contradicts itself within two sentences while also contradicting the ground truth. No legal citation provided.

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SELF-065 (EStG § 33 Abs. 1a)	Adjustment of income tax brackets for inflation (elimination of cold progression). (Cites no specific law)	Partly Correct. Correctly identifies the law's general purpose (mitigating progressive tax burden) but cites its own knowledge cutoff (October 2023) as a reason it cannot fully answer, then deflects to external sources instead of providing a direct answer. No legal citation provided.	Incorrect. Claims it limits tax liability for agriculture and self-employment income under § 10 Abs. 3 EStG, which has nothing to do with inflation adjustment or cold progression.	Fail. No relevant document found in the knowledge base - question left unanswered.

4.1 Quantitative Evaluation (BERTScore)

For the math part of the evaluation, I used BERTScore with baseline rescaling. I chose this because a regular BERTScore can be misleadingly high - since most models know basic German grammar, they get high points just for sounding human. By "rescaling," I'm basically subtracting the easy points so we can see the actual improvement each model provides.

The fine-tuned model achieved the strongest overall performance in terms of semantic similarity to the ground truth answers, with a BERTScore F1 of 0.2812, precision of 0.3236 and recall of 0.2437. These results indicate a high degree of alignment with reference answers in both content coverage and phrasing similarity.

The RAG model achieved an average BERTScore F1 of 0.2211, precision of 0.2706 and recall of 0.1764. While this demonstrates strong semantic similarity, the slightly reduced scores suggest that retrieval constraints and context selection introduce variability in output alignment compared to the fine-tuned model.

In contrast, the baseline inference model shows substantially lower BERTScore values (F1 = 0.1718, precision = 0.0537 and recall = 0.3087), indicating a conflict in lexical and structural similarity to the reference answers. However, this metric underestimates its actual performance, as shown in the qualitative evaluation.

4.2 Qualitative Evaluation

A manual review of 10 representative prompts was conducted to evaluate factual correctness and reasoning quality across the baseline, fine-tuned (SFT), and retrieval-augmented generation (RAG) models. The analysis reveals systematic failure modes for each approach.

4.2.1 Baseline Model (Inference)

The Inference model performed best overall in the qualitative spot-check, reaching a correct or partly correct conclusion in 8 out of 10 cases. However, its main weakness is that it over-explain. Instead of giving a short, precise legal answer, it produces long, conversational responses that often miss the specific paragraph or add conditions that are not in the ground truth. For example, in EMP-007 it cites the wrong paragraph (§ 16 Abs. 1 Z 8 EStG instead of § 26 EStG), and in GRESt-AT-015 it completely misses the joint liability rule. One hallucination was also observed - a non-existent § 34 Abs. 1 EStG in EstG-23-040.

This is also visible in the BERTScore results. Despite doing well qualitatively, the model scores the lowest F1 of 0.1718 (precision: 0.0537, recall: 0.3087). The very low precision but high recall suggests that the model covers the right topic but generates too much irrelevant content around it. This makes sense - the ground truth answers are very short and formal, while the model writes much longer explanations. So in this case, BERTScore is more of a measure of writing style than actual correctness, which is an important limitation to keep in mind when interpreting the results.

Based on the spot-check analysis, three error types were identified for the Inference model. The most common was Stylistic Verbosity (all 10 answers) where the model produces unnecessarily long responses instead of concise statutory answers. This was often combined with False Citation (citing a real paragraph that does not support the conclusion, e.g., GRESt-AT-015). In the most severe case (EstG-23-040), the model produced an Hallucination by referencing a non-existent legal provision. These error types suggest that while the Inference model has strong general legal knowledge, it lacks the precision needed for a reliable legal advisory tool.

4.2.2 Supervised Fine-Tuned (SFT) Model

The Fine-Tuned model shows the weakest qualitative performance, reaching a correct or partly correct conclusion in only 4 out of 10 cases. Its dominant failure mode is Terminological Drift - the model produces formally structured, concise responses that closely mirror the style of Austrian legal text, but frequently drifts into the wrong statutory category. In several cases the model correctly answers "No" but for entirely the wrong legal reason (EMP-007, EstG-23-040, SELF-065).

In more severe cases, Terminological Drift escalates into Hallucination, where the model not only cites the wrong law but fabricates legal reasoning entirely. For example, in SELF-023 it claims NoVA does not apply because "the business already exists".

Interestingly, despite its weak qualitative performance, the Fine-Tuned model achieves the highest BERTScore F1 of 0.2812. This is because fine-tuning successfully taught the model to mimic the formal, compressed style of Austrian legal text - making outputs look similar to the ground truth even when factually wrong.

4.2.3 Retrieval-Augmented Generation (RAG) Model

The RAG model reached a correct or partly correct conclusion in only 4 out of 10 cases, with 2 complete retrieval failures (TAX-INTL-053, SELF-065). Its dominant failure mode is Retrieval Sensitivity - its performance is connected to the quality and coverage of the knowledge base. When relevant documents are not retrieved, the model either returns no answer at all or misapplies retrieved content from an adjacent legal domain.

The most common error types observed were Retrieval Sensitivity combined with Hallucination. In EMP-007 it cites § 16d Abs. 1 Nummer 1 des österreichischen Steuergesetzes - an entirely fabricated reference. Despite being designed to reduce hallucination by grounding generation in retrieved source documents, the RAG model still produced hallucinated content in 3 out of 10 cases. This reflects a key design trade-off: while the model prioritises caution by returning no answer when retrieval fails, this safety mechanism only holds when the knowledge base gap is recognised. When retrieval returns partially relevant content from other documents in the knowledge base, the model still attempts to generate an answer and the result can be just as misleading as a hallucination.